

Gamma Knife Radiosurgery for Uveal Melanoma



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Keywords

- Uveal melanoma • Stereotactic radiosurgery • Gamma Knife
- Globe salvage therapy

Key points

- Uveal melanoma is the most common primary intraocular malignancy in adults.
- The management of uveal melanoma has developed a trend toward globe salvage therapies; Stereotactic Radiosurgery using Gamma Knife is one of the established methods.
- Gamma Knife has yielded promising results in terms of local tumor control; however, it is not free of complications.

INTRODUCTION

Uveal melanoma is the most common primary intraocular malignancy in adults [1]. Among these tumors, more than 90% arise from the choroid (Fig. 1), 5% to 8% arise from the ciliary body, and 3% to 5% arise from the iris [2].

The annual incidence of uveal melanoma is 0.2 to 6.3 cases per million population and shows significant racial differences [1,2]. Several predisposing factors have been reported, ranging from patient factors, including light skin color, light eye color, oculodermal melanocytosis, dysplastic nevus syndrome, iris nevus, and choroidal nevus, to environmental factors, such as UV light exposure [2].

Uveal melanomas are broadly grouped into small, medium, and large (Table 1), depending on the tumor dimensions [3]. The management of uveal

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